

Course Syllabus

This Syllabus contains a number of links or hyperlinks to other websites that may be more difficult to access from a mobile device. For optimal access to linked resources, please view this page in a standard Web browser. You can also use your operating system to [download/print this page and save this Syllabus as an accessible PDF](#)  [. \(https://www.wikihow.com/Convert-a-Webpage-to-PDF\)](https://www.wikihow.com/Convert-a-Webpage-to-PDF).

Course Information

Course Title: ALY3015 - Intermediate Statistics for Data Analysis

Course Number (CRN): 25079

Course Description::

This course expands upon the earlier introduced statistical approaches. Emphasizes more advanced analysis and multivariate methods. The goal is to provide students with the fundamental data management, review, reengineering, and exploration skills as necessary data analytical competencies..

Term: 202710

Start and End Dates: September 9 – December 20, 2026

Credit Hours: 3

Course Format: Online

Campus/Room: Online

Course Website: <https://northeastern.instructure.com>

Meeting Days/Times:

This course is 100% online. Your instructor will be available once a week if you have questions or need additional guidance. Times will be on Wednesdays at 6:00 PM EST. This time can change depending on instructor or students request. If changes happen, instructor will notify using The Announcements Section of the course's CANVAS page.

Instructor Information

Full Name:

Dr. Diana Chiluiza, PhD (she/hers).

Main communication with instructor will be handled using the INBOX option on our course's CANVAS page. This is located on the left column. Please use this as the main/primary form of communication.

NU Email Address: [dchiluiza@northeastern.edu \(mailto:dchiluiza@northeastern\)](mailto:dchiluiza@northeastern.edu)

Office Hours: *Request time with your instructor using the INBOX option in CANVAS course's page.*

Second Point of Contact:

If some concern about the course arises and is not addressed by the instructor or advisor, please reach out to the Second Point of Contact.

Second Point of Contact: Prof. Vladimir Shapiro, PhD.

Second Point of Contact Email Address: v.shapiro@northeastern.edu

Course Requirements

(https://rpubs.com/Dee_Chiluiza/home)

1. Microsoft Office (Excel, Word, PowerPoint)
2. R , install the latest version R (it's Free). Link: <https://www.r-project.org/> (<https://www.r-project.org/>) or <https://cran.r-project.org/bin/windows/base/> (<https://cran.r-project.org/bin/windows/base/>)
3. R Studio. install the latest Desktop version (it's Free). Link: <https://www.rstudio.com/products/rstudio/download/#download> (<https://www.rstudio.com/products/rstudio/download/#download>)

Find Canvas technical support and resources, including 24/7 phone (1-833-450-3937) and chat support, under the Help icon in Canvas. Access Northeastern Technical support by calling 617-373-4357 (xHELP) or emailing: help@northeastern.edu (<mailto:help@northeastern.edu>). To request IT support, visit the [Information Technology Services \(ITS\) Support Portal](https://service.northeastern.edu/tech)  (<https://service.northeastern.edu/tech>).

ALY2010

Good knowledge of Excel and Power Point.

1. Elementary Statistics, A Step by Step Approach: A Brief Version, 10th Edition, Allan G. Bluman, McGraw Hill Publishing, ISBN: 97812600420009
2. R in Action 2nd Edition. R. Kabacoff, Manning Publishing: ISBN 9781617291388
3. D. Chiliza. 2022. Introduction to analytics using R, R Studio and R Markdown Short manual series. https://rpubs.com/Dee_Chiluiza/home (https://rpubs.com/Dee_Chiluiza/home)

Course Expectations

PLO1: Examine major concepts and skills in statistical and mathematical analysis related to data analytics.

PLO2: Apply the principles, tools and methods of analytics to a project within a sponsoring organization to assist with the extraction, development, delivery, translation, implementation of data analysis for tactical or strategic decision-making in organizations, and successfully communicate data-driven insights.

PLO3: Complete a substantial project that requires the application of the principles, tools and methods of analytics systems technology to a comprehensive real-world problem or project related to data analyses for tactical and/or strategic decision making

PLO4: Develop and justify effective paths to resolution of an analytical problem that may be complicated by the competitive environment, opposing interests, divergent or uncertain data and information.

PLO5: Investigate major theories, tools, and approaches in data analytics to identify and successfully communicate data-driven insights for informed decision-making.

PLO6: Articulate and defend the significance and implications of the work in data analytics in terms of challenges and trends in a local, national or global context in relation to data governance, policy, or ethics.

Based on satisfactory completion of this course, a student should be able to:

CLO1: Communicate data information and insights using data visualizations such as graphs, dashboards, scorecards, spatial data representations, pictograms and special notations.

CLO2: Evaluate and create appropriate visualizations for a specified audience using key informational design concepts and storytelling techniques.

CLO3: Analyze the ethical considerations and diverse perspectives within an organization to communicate relevant and accurate data.

CLO4: Apply strategies for systematic visual exploration of multidimensional and heterogeneous data sets.

CLO5: Implement effective visualizations using R.

One (1) academic credit requires 50 minutes a week of classroom or faculty instruction and about two hours of out-of-class student work for a 15-week course; 65 minutes a week of classroom or direct faculty instruction and about 2.5 hours of out-of-class student work for a 12-week course; 100 minutes a week of classroom or direct faculty instruction and about 3.5 hours of out of class student work for a 7.5-week course.

For online three-credit course, 15-week course, students should expect a minimum of 5 hours of out-of-class work. For 12-weeks courses a minimum of 7.5 hours of out of class student work for a 12-week course.

To facilitate interaction, students are expected to review the online postings on a regular basis even after they have posted their own minimum required postings. The University seeks to provide a supportive environment that is conducive to learning, the pursuit of truth, the exchange of knowledge, the intellectual development of students, and the general good of society. Learn more about the [Code of Student Conduct | Office of Student Conduct and Conflict Resolution \(northeastern.edu\)](https://osccr.sites.northeastern.edu/code-of-student-conduct/)  (<https://osccr.sites.northeastern.edu/code-of-student-conduct/>).

This is an online course, attendance to office hours are not required.

Students need to read the weekly learning material presented by the instructor, including lectures, videos, book chapters, etc.

Students must prepare and submit weekly or biweekly assignments, observing their corresponding deadlines.

- **Deadlines:** All assignments are due by the specified date and time.
- **Late Submissions (Up to 1 Week):** Assignments submitted within seven days after the deadline will incur a **10% grade reduction penalty**.
- **Final Cutoff:** Submissions close permanently **one week after the deadline**. Assignments will not be accepted after this grace period.

Permission to make up work may be granted by instructors for reasonable cause, per CPS [Attendance Requirements](https://catalog.northeastern.edu/professional-studies/academic-policies-procedures/attendance-requirements/) , or by [Requesting an](https://catalog.northeastern.edu/professional-studies/academic-policies-procedures/attendance-requirements/)

Incomplete Grade [↗\(https://catalog.northeastern.edu/professional-studies/academic-policies-procedures/incomplete-grade-policy/\)](https://catalog.northeastern.edu/professional-studies/academic-policies-procedures/incomplete-grade-policy/).

GenerativeAI (GenAI) tools can support learning when used thoughtfully, but they can also produce inaccurate, biased, or fabricated information. You are responsible for critically evaluating all AI-generated content for accuracy, quality, and integrity. Using AI in ways that are not allowed by your instructor, submitting unauthorized AI-generated content as if you created it, or relying on AI to complete substantial portions of an assignment, may violate academic integrity policies.

Please refer to specific assignment guidelines and ask your instructor to understand if or how you can use GenAI. Depending on the assignment, GenAI may be allowed, restricted, or prohibited. Remember that GenAI should never replace your own thinking, analysis, or voice, and your submitted work should be consistent with your abilities. Because AI tools can interfere with skill development, especially in writing, critical thinking, language learning and math comprehension, overreliance may negatively impact your learning. Your instructor may ask you to disclose and cite your use of GenAI.

You are also expected to use AI tools in ways that respect intellectual property, privacy, and university policies. Do not upload course materials, assignment prompts, other students' work, or sensitive information to tools unless consistent with university policy and permitted by your instructor. Do not enter confidential university information, personal information, research data, or other protected content into AI tools unless the tool has been approved by Northeastern University for that explicit use. Where applicable, you should use university-approved AI tools and comply with institutional technology and data security requirements.

Course-Specific Guidelines

The following expectations apply specifically to this course:

- **Permitted vs. Restricted Assignments:** AI tools (e.g., ChatGPT) are permitted **only for brainstorming, outlining, and initial proofreading** on all major essays. AI generation of raw text, code, or direct answers for quizzes, exams, and short reading responses is **strictly prohibited**.
- **Disclosure Requirement:** If you use AI during your preparation, you must include an **AI Disclosure Statement** at the end of your submission detailing:
 1. The tool(s) used.
 2. The exact prompt(s) entered.
 3. A brief explanation of how you integrated or modified the tool's output into your own work.

- **Approved & Prohibited Tools:**

- *Approved:* Institutional AI assistants, Grammarly (spelling/grammar checking only), Gemini/DeepSeek/ChatGPT/Claude for ideation.
- *Prohibited:* Generating automated text to bypass writing assignments, as well as using AI to generate code or solve problem sets.

- **Course Consequences:** Unauthorized AI use or failure to provide a complete disclosure statement will result in a **zero (0) on the assignment** and a referral to the Honor Council for academic dishonesty.

Failure to follow these guidelines, including misuse or non-disclosure of AI, will be treated as a violation of academic integrity and addressed on a case by case basis according to the [Northeastern University AI policy](https://policies.northeastern.edu/policy125/)  (<https://policies.northeastern.edu/policy125/>) and to [CPS academic integrity policies](https://cps.northeastern.edu/current-students/academic-integrity/).  (<https://cps.northeastern.edu/current-students/academic-integrity/>)

Course Methodology

Add assignments here

Activity 1:

Activity 2:

Activity 3:

Week/Dates	Topic	Material to Complete	Exercises/Assignment with Due Dates

Evaluation Standards

94-100%	A
90-93.9%	A-
87-89.9%	B+
84-86.9%	B
80-83.9%	B-
77-79.9%	C+
74-76.9%	C

70-73.9%	C-
67-69.99	D+
64-66.99	D
60-63.99	D-
59.9% or below	F

Example:

Course Participation: 20%

Weekly Assignments: 40%

Case Study: 20%

Final Exam: 20%

Total: 100%

Per **CPS Academic Appeals Process and Procedures** [↗\(https://catalog.northeastern.edu/professional-studies/academic-policies-procedures/student-appeals/\)](https://catalog.northeastern.edu/professional-studies/academic-policies-procedures/student-appeals/), in all cases, students are encouraged to informally discuss concerns with the faculty member who taught the course to see if it is possible to reach an agreement on the issue(s). If the student is not satisfied with the outcome of this discussion, or if the student is not comfortable discussing the issue with the instructor, the student should request a meeting with their academic and career advisor who can help facilitate the process of resolving the issue(s).

Student Well-Being

All students have access to We Care – 617-373-7591 wecare@northeastern.edu for health concerns, referrals for mental health and assistance with leaves of absence. Additionally we have FIND@ Northeastern - Available 24/7/365 at 1-877-233-9477 (domestic) 1-781-457-7777 (international) <https://uhcs.northeastern.edu/findnortheastern/> ➞ <https://uhcs.northeastern.edu/findnortheastern/>

Library Services

The Northeastern University Library includes over 900,000 print volumes, 206,500 e-books, and 70,225 electronic journals. For more information, visit <http://library.northeastern.edu/> ➞ <http://library.northeastern.edu/>.

Tutoring Service

Northeastern University's College of Professional Studies offers the following tutoring options:

- Twenty-four hours per academic year of tutoring from Wyzant to help ensure your success in challenging courses.
- 45-minute writing appointments at the [Writing Center](#) ➞ https://cssh.northeastern.edu/writingcenter/#_ga=2.183028835.470613829.1693925134-84983876.1644263066 cover writing topics from start to finish such as research, organization, development, introductions, conclusions, multimedia projects, and formatting.
- 50-minute appointments at the [International Tutoring Center](#) ➞ <https://international.northeastern.edu/itc/tutoring/> for English language instruction, including conversation, writing, reading, and career preparation. Professional ESL tutors offer various formats of tutorials, including in-person, online, one-on-one, and group appointments.

To learn more about tutoring, visit [Tutoring Services](#) ➞ <https://cps.northeastern.edu/current-students/tutoring-services/>.

Institutional Policies

End-of-Course Evaluation Surveys

At the end of this course, you will receive an email requesting that you complete a confidential [Student Evaluation of the course](https://catalog.northeastern.edu/professional-studies/academic-policies-procedures/evaluationkit/)  (<https://catalog.northeastern.edu/professional-studies/academic-policies-procedures/evaluationkit/>). Student evaluation data is integral to assessing academic quality and student experience at CPS, and all students are strongly encouraged to complete them, each term.

Academic Integrity

Independent and original scholarship, alongside proper attribution, helps ensure that students derive the most from their educational experience. Dishonesty, shortcuts, and/or carelessness disregards the fundamental values of our community and will be reported via the Faculty Advisor Communication Tool (FACT) and to the Office of Student Conduct and Conflict Resolution (OSCCR). Your course may use systems to help ensure the authenticity of assessments, such as TurnItIn and exam proctoring. Work displaying apparent academic integrity lapses may earn a failing grade, a zero, or result in a failing grade for the class. Students may not withdraw from a class to avoid a failing grade due to lapses.

As members of the academic community, students must become familiar with their rights and responsibilities. Review the university's policy on Academic Integrity here: [Academic Integrity Policy | Office of Student Conduct and Conflict Resolution \(northeastern.edu\)](https://osccr.sites.northeastern.edu/academic-integrity-policy/#_ga=2.190686318.470613829.1693925134-84983876.1644263066)  (https://osccr.sites.northeastern.edu/academic-integrity-policy/#_ga=2.190686318.470613829.1693925134-84983876.1644263066). Students can find resources to assist in understanding and meeting these expectations by visiting [Academic Integrity | Northeastern University College of Professional Studies](https://cps.northeastern.edu/current-students/academic-integrity/)  (<https://cps.northeastern.edu/current-students/academic-integrity/>).

Student Accommodations

The College of Professional Studies is committed to providing equitable access to learning opportunities to students with documented disabilities (e.g. mental health, attentional, learning, chronic health, sensory, or physical). Please contact [The Disability Access Services](https://disabilityaccessservices.sites.northeastern.edu/)  (<https://disabilityaccessservices.sites.northeastern.edu/>) to engage in a confidential conversation about the process for requesting reasonable accommodations in the classroom and clinical or lab settings. Accommodations are not provided retroactively so students are encouraged to register with the DAS as soon as they begin their program and access their resources throughout their learning journey.

At the end of this course, you will receive an email requesting that you complete a confidential [Student Evaluation of the course](https://catalog.northeastern.edu/professional-studies/academic-policies-procedures/evaluationkit/) [\(https://catalog.northeastern.edu/professional-studies/academic-policies-procedures/evaluationkit/\)](https://catalog.northeastern.edu/professional-studies/academic-policies-procedures/evaluationkit/). Student evaluation data is integral to assessing academic quality and student experience at CPS, and all students are strongly encouraged to complete them, each term.

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Northeastern University and the Office of Institutional Diversity and Inclusion (OIDI) are committed to equal opportunity, diversity, and social justice while building a climate of inclusion on and beyond campus. In the classroom, members of the University community work to cultivate an inclusive environment that denounces discrimination through innovation,

collaboration, and an awareness of global perspectives on social justice. Visit <https://belonging.northeastern.edu>  (<https://belonging.northeastern.edu>) for complete information on Diversity and Inclusion.

Northeastern University and its faculty are committed to creating a safe and open learning environment for all students. If you or someone you know has experienced discrimination (including discrimination based on sex, gender, gender identity, gender expression, sexual orientation, pregnancy or pregnancy related condition, race, religion, national origin, disability status, veteran status etc.), or sexual violence (including sexual harassment, sexual assault, dating/domestic violence, or stalking), please know that help and support are available. Northeastern strongly encourages all members of the community to take action, seek support, and report incidents of discrimination, harassment, and sexual violence to the Office for University Equity and Compliance (OUEC) through the Online Discrimination Complaint Form found at <https://www.northeastern.edu/ouec/file-a-complaint/>  (<https://www.northeastern.edu/ouec/file-a-complaint/>).

Please be aware that faculty members are Mandatory University Reporters who are required to disclose information about alleged discrimination, harassment, and sexual violence to the OUEC. If the OUEC receives a report, a member of their office will reach out to offer information about available rights, support resources and pathways towards a resolution as a member of the campus community. Community members are not required to respond to this outreach.

If you, or another community member you know wishes to speak to a confidential resource who does not have this reporting responsibility, please contact any of the following confidential resources. These confidential resources are not required to report allegations of discrimination to the University without your signed release.

- **Find@Northeastern**  (<https://uhcs.northeastern.edu/findnortheastern/>): Offers 24/7 mental health support via phone at 877.233.9477 (in the U.S.) or +1.781.457.7777 (outside the U.S.).
- **Sexual Violence Resource Center**  (<https://open.studentlife.northeastern.edu/meet/sexual-violence-resource-center/>): The SVRC provides confidential, trauma-informed support services to Northeastern students who have experienced any form of sexual violence (i.e., sexual assault, sexual harassment, sexual exploitation, domestic/dating violence, and/or stalking). Request services online at bit.ly/svrequestform  (https://neu.co1.qualtrics.com/jfe/form/SV_3WzpBPgGQkhwUYK).
- **Confidential Resource Advisor:**  (<https://open.studentlife.northeastern.edu/meet/confidential-resource-advising/>) The CRA provides confidential, restorative informed support services to Northeastern students who have been accused of sexual or identity-based harm. Request services online at bit.ly/svrequestform  (<https://bit.ly/svrequestform>).

Pregnant / parenting students: Please know that the OUEC, housing the University's Title IX Coordinator, can work with students who are pregnant and/or parenting to ensure they have equal access to education programs and activities. For additional support, please contact OUEC (ouec@northeastern.edu (<mailto:ouec@northeastern.edu>)).

Please visit www.ouec.northeastern.edu  (<http://www.ouec.northeastern.edu/>) for a complete list of [reporting options](https://ouec.northeastern.edu/reporting-options/)  (<https://ouec.northeastern.edu/reporting-options/>) and [support resources](https://ouec.northeastern.edu/resources/)  (<https://ouec.northeastern.edu/resources/>) both on- and off-campus and contact the OUEC (ouec@northeastern.edu) at any time.

Northeastern University Copyright Statement

The instructor reserves the right to amend this syllabus, both online and in the document itself, during the term and will notify students of the change(s). The revised syllabus is the official record of class policies and the schedule of due dates.

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Course Summary:

Date	Details	Due
Sun Sep 13, 2026	 Module 1 Discussion: Introduce Yourself (https://northeastern.instructure.com/courses/271875/assignments/3391392)	due by 11:59pm
Fri Sep 18, 2026	 Module 2 Discussion (https://northeastern.instructure.com/courses/271875/assignments/3391390)	due by 11:59pm
Fri Sep 25, 2026	 Module 3 Discussion (https://northeastern.instructure.com/courses/271875/assignments/3391384)	due by 11:59pm
Fri Oct 2, 2026	 Module 4 Discussion (https://northeastern.instructure.com/courses/271875/assignments/3391383)	due by 11:59pm

Date	Details	Due
Fri Oct 9, 2026	 <u>Module 5 Discussion</u> https://northeastern.instructure.com/courses/271875/assignments/3391391	due by 11:59pm
Fri Oct 16, 2026	 <u>Module 6 Discussion</u> https://northeastern.instructure.com/courses/271875/assignments/3391381	due by 11:59pm
Fri Oct 23, 2026	 <u>Module 7 Discussion</u> https://northeastern.instructure.com/courses/271875/assignments/3391389	due by 11:59pm
Fri Oct 30, 2026	 <u>Module 8 Discussion</u> https://northeastern.instructure.com/courses/271875/assignments/3391388	due by 11:59pm
Fri Nov 6, 2026	 <u>Module 9 Discussion</u> https://northeastern.instructure.com/courses/271875/assignments/3391382	due by 11:59pm
Fri Nov 13, 2026	 <u>Module 10 Discussion</u> https://northeastern.instructure.com/courses/271875/assignments/3391387	due by 11:59pm
Fri Nov 20, 2026	 <u>Module 11 Discussion</u> https://northeastern.instructure.com/courses/271875/assignments/3391385	due by 11:59pm
Fri Dec 4, 2026	 <u>Module 12 Discussion</u> https://northeastern.instructure.com/courses/271875/assignments/3391386	due by 11:59pm